12

Nmap to discover Live Hosts.

Aim:

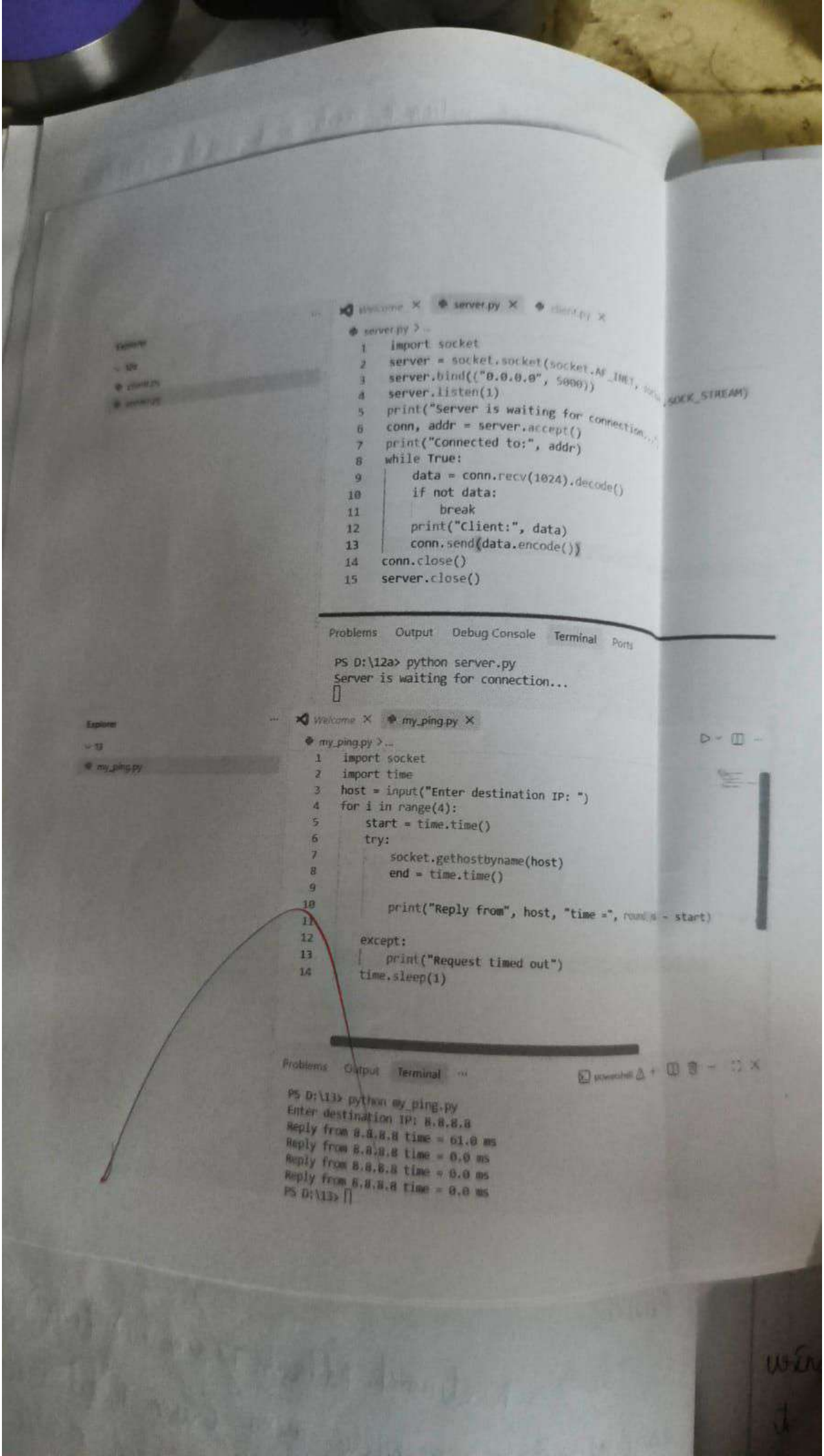
To discover live hosts in a network using Nmap by performing ARP scan, ICMP scan, TCP Ping scan, and UDP Ping scan on the Try Hack Me Platform.

Procedure:

* Log in to the Try Hack Me Platform.
* Open the Nmap room and start the target machine.
* Start the Attack box and open the terminal.
* Enter the target machine's IP address.
* Perform ARP, ICMP, TCP, and UDP scans using Nmap commands.
* Observe and record the live hosts discovered.

Result:

Thus, live hosts in the network were discovered succes using Nmap, with ARP, ICMP, TCP, and UDP scanning techniques.



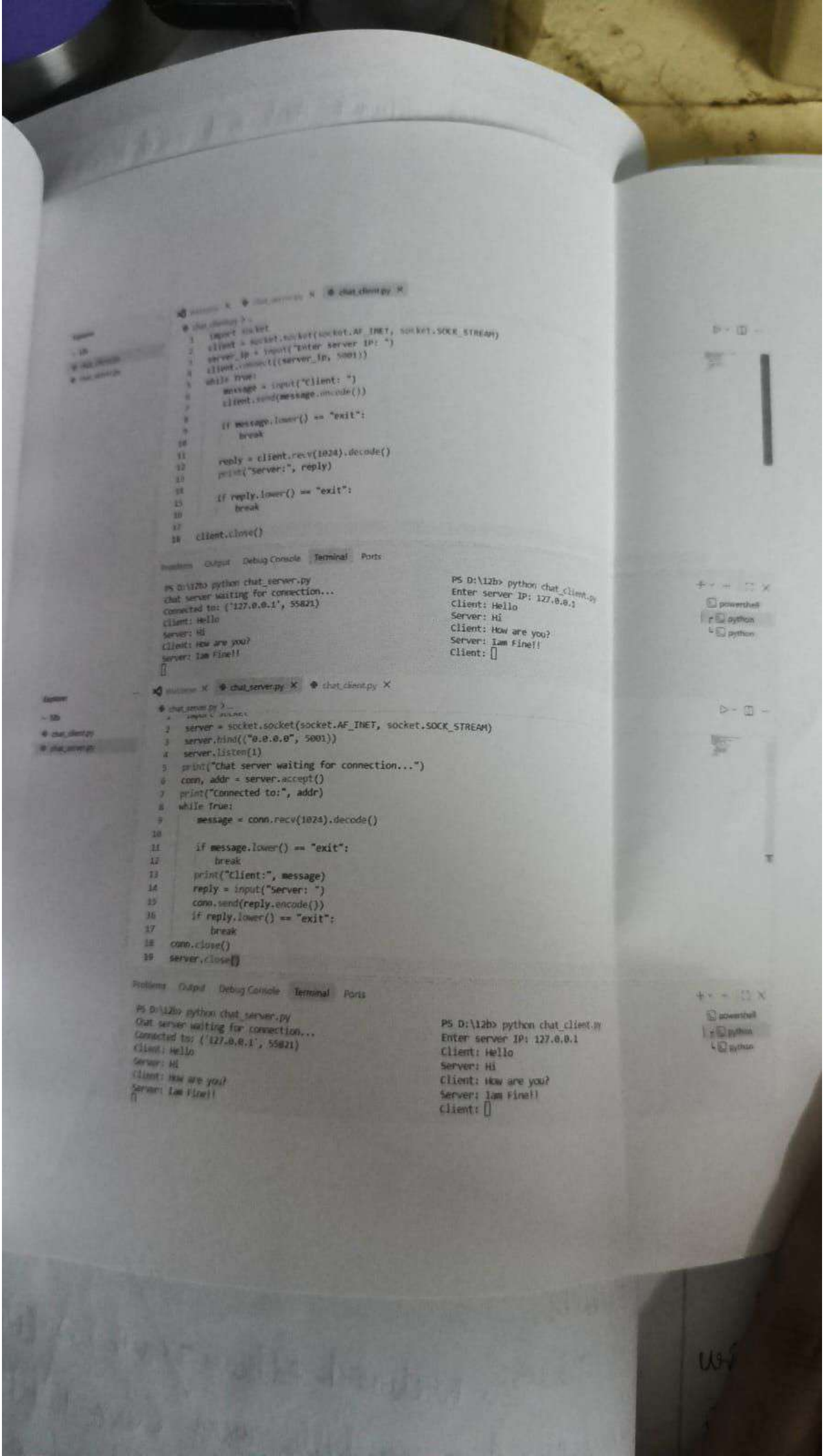
```python
import socket
server = socket.socket(socket.AF_INET, SOCK_STREAM)
server.bind(("0.0.0.0", 5000))
server.listen(1)
print("Server is waiting for connection...")
conn, addr = server.accept()
print("Connected to:", addr)
while True:
    data = conn.recv(1024).decode()
    if not data:
        break
    print("Client:", data)
    conn.send(data.encode())
conn.close()
server.close()
```

```
PS D:\12a> python server.py
Server is waiting for connection...
```

```python
import socket
import time
host = input("Enter destination IP: ")
for i in range(4):
    start = time.time()
    try:
        socket.gethostbyname(host)
        end = time.time()

        print("Reply from", host, "time =", round(s - start)
    except:
        print("Request timed out")
    time.sleep(1)
```

```
PS D:\13> python my_ping.py
Enter destination IP: 8.8.8.8
Reply from 8.8.8.8 time = 61.0 ms
Reply from 8.8.8.8 time = 0.0 ms
Reply from 8.8.8.8 time = 0.0 ms
Reply from 8.8.8.8 time = 0.0 ms
PS D:\13>
```



```python
import socket
client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server_ip = input("Enter server IP: ")
client.connect((server_ip, 5001))
while True:
    message = input("Client: ")
    client.send(message.encode())

    if message.lower() == "exit":
        break

    reply = client.recv(1024).decode()
    print("Server:", reply)

    if reply.lower() == "exit":
        break

client.close()
```

```
PS D:\12b> python chat_server.py
Chat server waiting for connection...
Connected to: ('127.0.0.1', 55821)
Client: Hello
Server: Hi
Client: How are you?
Server: Iam Fine!!
```

```
PS D:\12b> python chat_client.py
Enter server IP: 127.0.0.1
Client: Hello
Server: Hi
Client: How are you?
Server: Iam Fine!!
Client:
```

```python
server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind(("0.0.0.0", 5001))
server.listen(1)
print("Chat server waiting for connection...")
conn, addr = server.accept()
print("Connected to:", addr)
while True:
    message = conn.recv(1024).decode()

    if message.lower() == "exit":
        break
    print("Client:", message)
    reply = input("Server: ")
    conn.send(reply.encode())
    if reply.lower() == "exit":
        break
conn.close()
server.close()
```

```
PS D:\12b> python chat_server.py
Chat server waiting for connection...
Connected to: ('127.0.0.1', 55821)
Client: Hello
Server: Hi
Client: How are you?
Server: Iam Fine!!
```

```
PS D:\12b> python chat_client.py
Enter server IP: 127.0.0.1
Client: Hello
Server: Hi
Client: How are you?
Server: Iam Fine!!
Client:
```